

Assaad El Halabi

Backend and DevOps Engineer at Oreyeon

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Experience:

Software Engineer III, Oreyeon

07/2022 – Present

- **Architected containerized, fault-tolerant backend services and RESTful APIs** on AWS Lambda & API Gateway—delivering 99.9% SLA-compliant, real-time batch metadata insights and slashing time-to-insight by 85% for airport operations teams.
- **Automated 90%+ of manual batch-handling tasks** (bulk creation, editing, visualization) via a complete web application with reusable Node.js/Python microservices driven by cross-functional team requirements, reducing operational overhead by up to 80 hours per month.
- **Built CI/CD pipelines** with GitHub Actions and AWS CodePipeline/CodeBuild—including automated testing, linting, and deployment—cutting deployment lead time by 60%, boosting code-quality metrics and uphold CMMC Level 1 and Level 2 compliance requirements.
- **Optimized Company Processes** by engineering strategic solutions such as an annotation management dashboard that slashed handler assignment from 20 minutes to 35 seconds, payment calculation from over 1 hour to under 39 seconds, and batch outcome reviews from 10 minutes to under 10 seconds—dramatically accelerating data science, data handling, and operations workflows.
- **Collaborated with product, ML, and ops teams** to enforce 99.9% uptime SLAs for time-sensitive airport applications, achieving CMMC Level 1 and Level 2 certifications, ensuring the integrity of Federal Contract Information (FCI) and Controlled Unclassified Information (CUI).
- **Modernized core infrastructure** using TypeScript, Docker, AWS SDK, ensuring scalable, observable, and maintainable pipelines for high-volume image processing.

Software Engineer, AUB

01/2021 – 06/2021

- Worked as part of a team of software, data engineers and analysts with two professors as part of the initiative [Khaddit Beirut](#).
- Automating data pre-processing and visualization pipelines and tasks using Python.
- Collaborating with team members and professors to create and customize an interface and dashboard to interact and visualize Beirut explosion data and insights.
- Detecting demographic patterns of the explosion aftermath aimed at guiding future research.

Technical Skills:

- **Back-End:** Node.js, Spring boot, AWS Lambda, AWS EC2, AWS ECR, Django.
- **Databases:** MySQL, PostgreSQL, SQL Server, AWS RDS, relational and non-relational databases.
- **Programming:** Python, Java, Javascript, Typescript, C++, Bash Scripting, OOP.
- **DevOps:** CI/CD pipelines, Agile Methodologies, Git, Linux
- **Cloud:** AWS Cloud Services, AWS Lambda, Docker, Docker Compose, Virtualization
- **Front-End Web:** HTML, CSS, JavaScript, React.js
- **Tools:** Github, Bitbucket, SVN, Jira, Azure DevOps, Github Actions.

Education:

OnRamp Lebanon - Fast-Track to Senior Bootcamp 04/2024 – Present

- Participating in intensive bootcamp accelerating career progression to senior-level engineering roles.
- Focus on advanced software engineering practices, system design, and leadership skills.

American University of Beirut

09/2021 – 02/2024

- **MS in Computer Science**
- **Full scholarship** covered by [MEPI](#).
- Peer-reviewed multiple revisions of a publication that has been published on [ACM Transactions on Asian and Low-Resource Language Information Processing](#).
- **Thesis:** Developed Arabic Conversational models powered by meta-learning and paraphrase-augmentation with little data as part of my master's thesis. Used Python, Google Colab, background jobs for automating data splitting and augmenting the partitions.

American University of Beirut

09/2018 – 06/2021

- **BS in Computer Science**
- **Full scholarship** covered by [USAID](#).
- **Dean's Honor List**, which recognizes students for their outstanding academic achievements.
- **Courses:** Advanced Algorithms and Data Structures, Software Engineering, Database Systems, Machine Learning, Data Science, Microeconomics.

Projects:

[JSColab](#) (04/2023)

Zero-setup notebook-based JavaScript environment for fast prototyping, inspired by Google colab, but for JavaScript and React.

- Fully featured JavaScript notebook code cells for fast prototyping and markdown cells for documentation.
- Ability to import any npm library or CSS library (e.g., Bulma) dynamically.

[CollaBoard](#) (05/2024)

Collaborative Whiteboard supporting real-time interaction among users for creative and planning purposes. OnRamp Bootcamp's final Project.

[AraDialAug](#): Arabic Chatbots with little data (02/2024)

Master's Thesis: Arabic Conversational models powered by meta-learning and paraphrase-augmentation with little data.

[WeTube](#) (11/2021 - 12/2021)

YouTube Clone using ReactJS and ASP.NET Core.

[Emotion and Face pose Classifier](#) (01/2021 - 04/2021)

Predicts emotion and if you are looking at the camera.

Languages:

Arabic (Native) | English (Proficient) | Portuguese (Intermediate) | Spanish (Intermediate)